field notes, entry no. 1, PyCon

Is there really *only one?*

Soul mates, measured with PyMC.

four datasets and one honest guess

your name

field notes, entry no. 2

One person, made for you.
Somewhere on Earth.

Randall Munroe asked it in *What If?*: what if everyone had exactly one soul mate, about your age, anywhere in the world?

How likely is it that you ever meet?
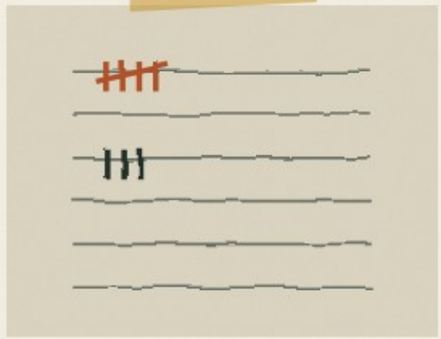


the premise

one dot, 8 billion people

so let us measure it.

field notes, entry no. 3

Four datasets and *one honest guess.*



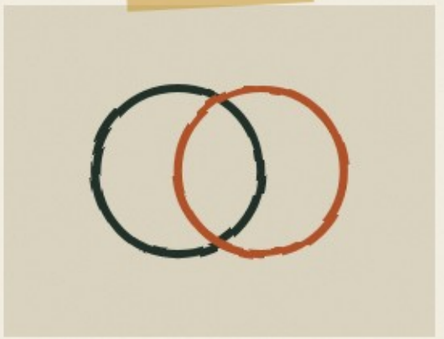
contact diaries

who you meet: 9 countries



UN population

how many are your age



HCMST 2017

who finds a partner



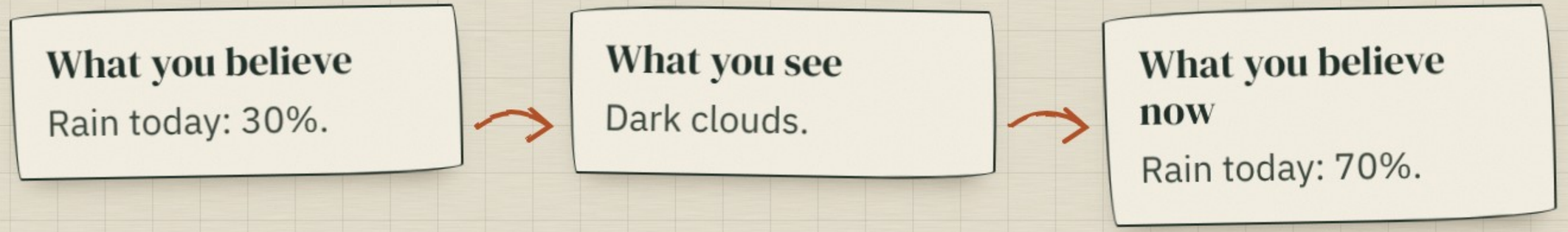
your guess

would you both know?

POLYMOD / France 2012 / UN WPP 2024 / HCMST, with PyMC, ArviZ, PreliZ and marimo

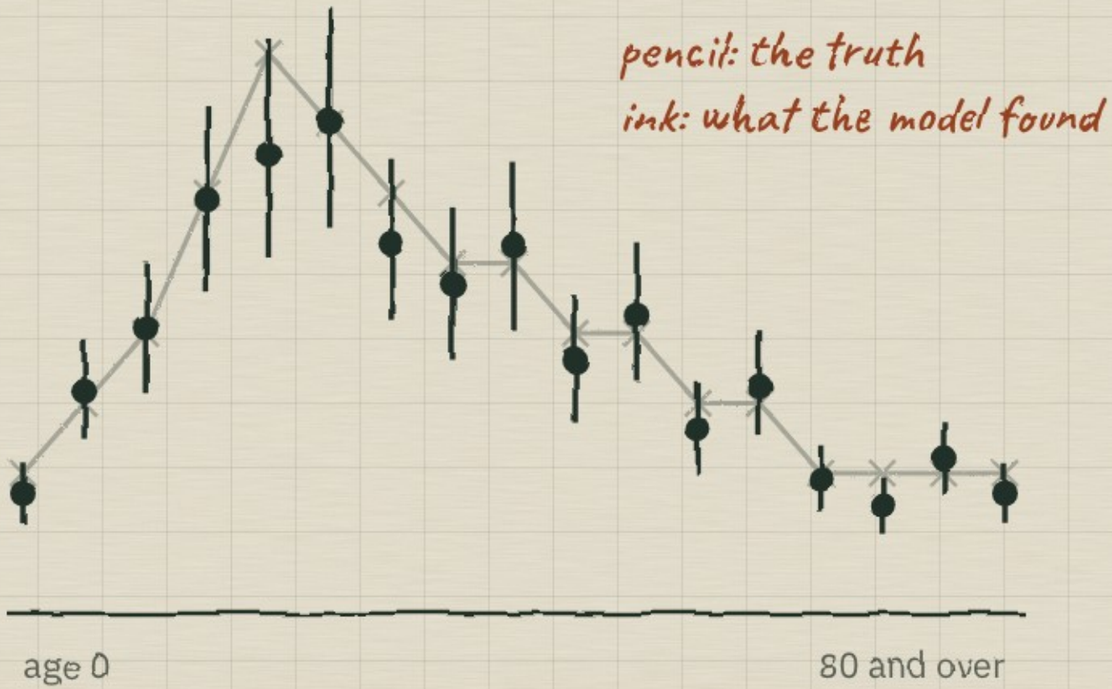
field notes, entry no. 4

Bayes is a forecast *you keep correcting.*



PyMC does the correcting. Every number tonight comes with a range of plausible values.

First, test it *on fake lives.*



We invent 6,000 diaries where we choose the truth.

Then the model has to find it again.

rule one: never fit real data first.

field notes, entry no. 6

People wrote down *everyone they met.*

11,107

diary days, in 9 countries

3 words

A contact means you talked, or touched.

not "locked eyes". met and talked.

new people of your own age per day, by age band



field notes, entry no. 7

Only people *your age* count.

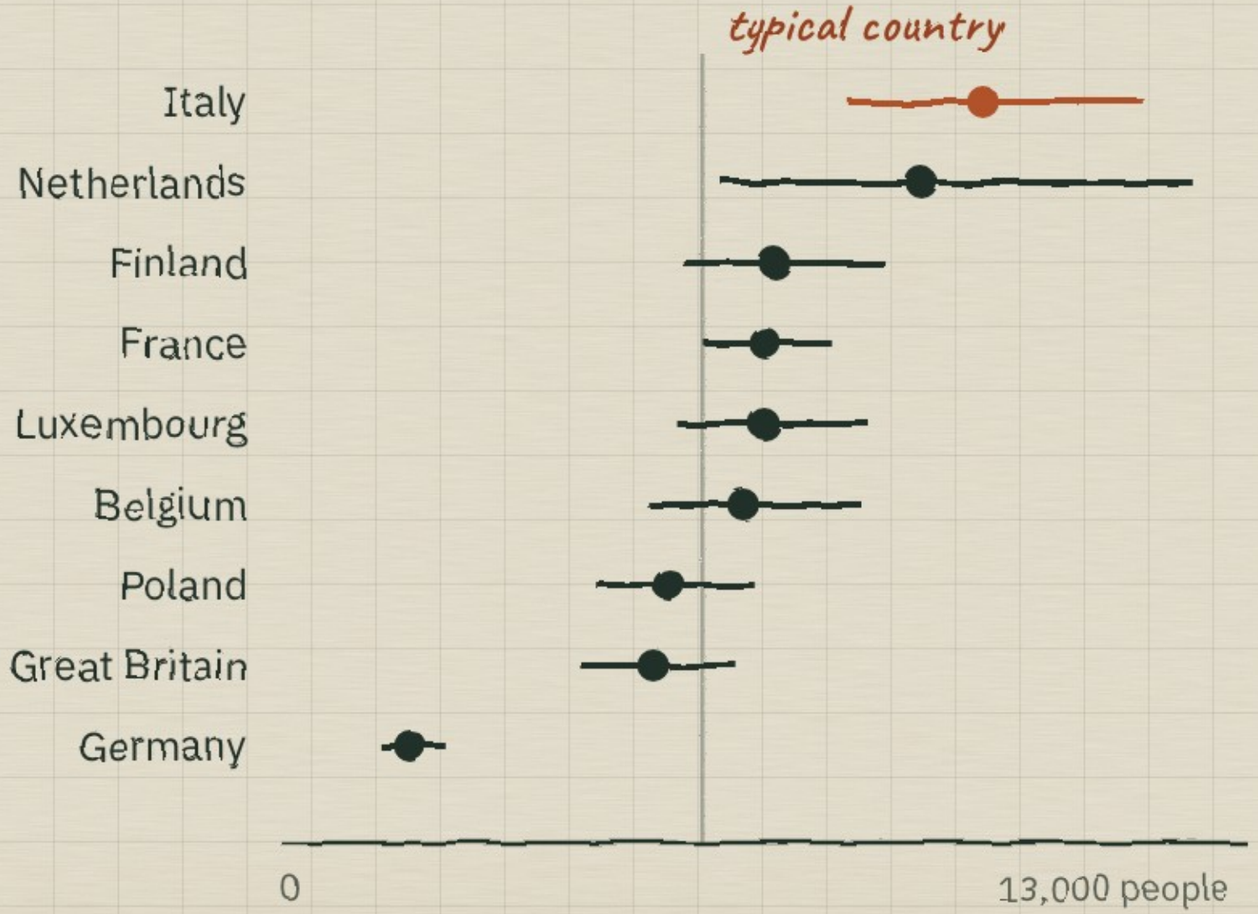
Your soul mate was born within 5 years of you. So were only **1 in 4** of the new people you meet.



this one fix changed everything.

field notes, entry no. 8

Some countries *talk more.*



New people of your age you talk to between 15 and 60.

Italy: 9,617
Germany: 1,777

culture, or how the survey asked? we cannot tell.

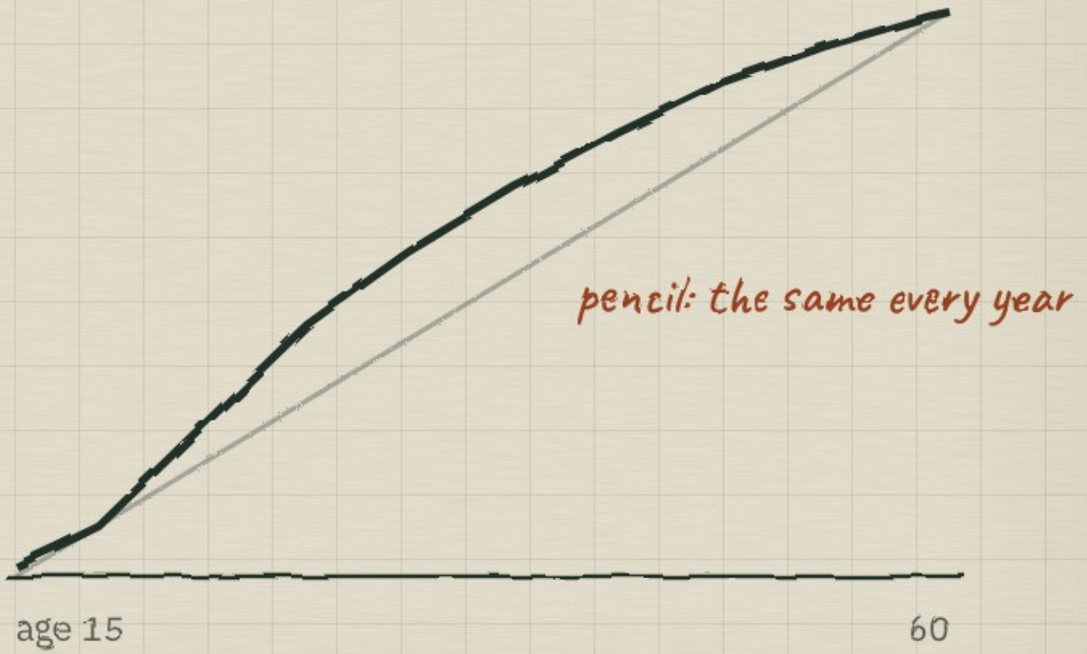
field notes, entry no. 9

about 5,800

people your age you will meet and talk to, between 15 and 60.

Somewhere in the world, that is from **2,000** to **17,000**.

most of them before you turn 35.



field notes, entry no. 10, chapter three: the odds

Your pool:
1.33 billion people.

Everyone alive aged 25 to 35, if you are 30.

UN World Population Prospects 2024



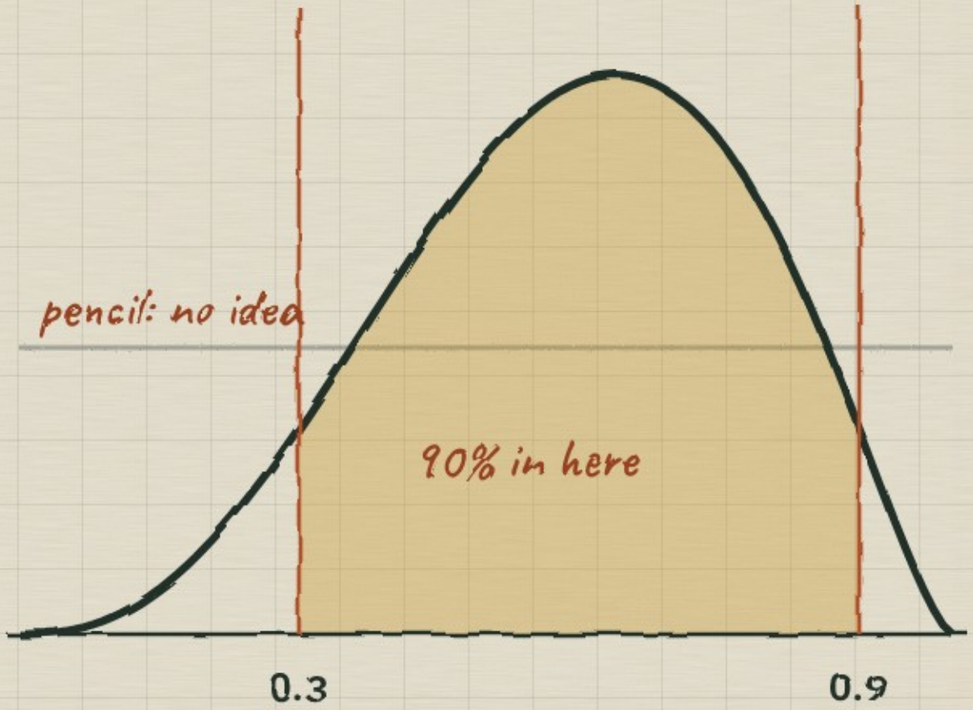
2,400 dots, each about 550,000 people

field notes, entry no. 11

The only number *nobody measured.*

"I am 90% sure we would both know, with a chance between 0.3 and 0.9."

PreliZ turns the sentence into the flattest curve that fits it.



field notes, entry no. 12

1 in 325,800

the chance to find your one soul mate, somewhere on Earth.

94% interval: 1 in 1,500,000 to 1 in 115,000

Still, about 4,100 people of your generation do find theirs.

field notes, entry no. 13

And if you are *lucky?*

BEST CASE

You live in **Italy**, the most talkative country.

1 in 231,900

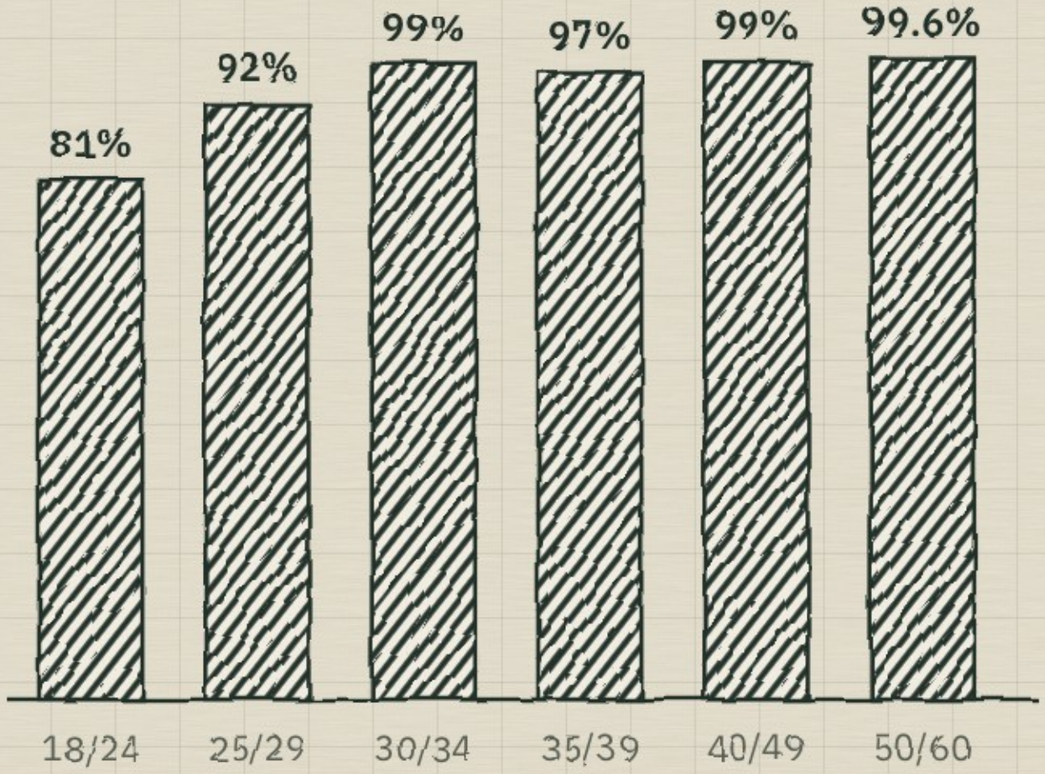
And you **always** recognize each other.

1 in 138,200

with exactly one soul mate, it does not get better than this.

field notes, entry no. 14, chapter four: the B side

But almost everyone *finds someone.*



share who ever had a partner, by age

96%

of people aged 18 to 60 have had a partner.

How Couples Meet and Stay Together, 2017, US

field notes, entry no. 15

Some of us talk to *many strangers.* Some to few.

Give every person their own number of contacts, and the model fits the young.

At ages 18 to 24 the data says **81%** and the model **78%**. Before, it said 68%.

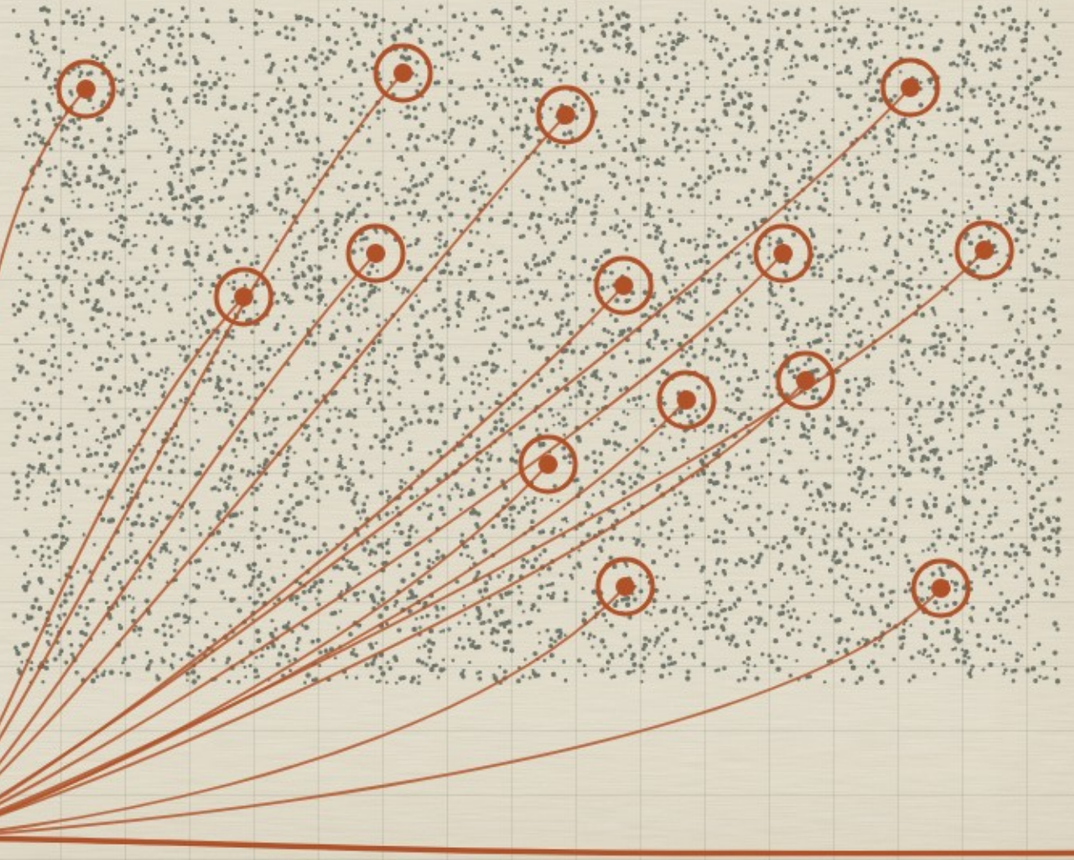


field notes, entry no. 16

You do not have one.

You meet about 14.

About **1 in 424** of the people your age you talk to could become a partner.



5,783 dots: the people your age you will talk to. Rust: the 14 who could be the one.

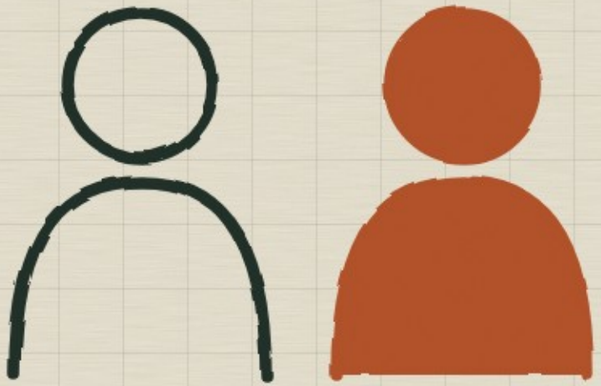
field notes, entry no. 17

In a room *built for it:*

1 in 6

Four minutes of talking. In 1 of 6 dates, both said yes.

Columbia speed dating study: 4,189 dates, 551 students. Fisman, Iyengar, Kamenica and Simonson, 2006.



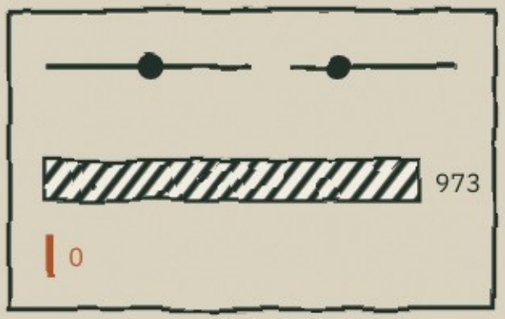
students who signed up for dating. not the street.

field notes, entry no. 18

Soul Mate Roulette, *live.*

Spin 1,000 lives. Pick your age, your country and your own guess.

```
uv run marimo run demo/roulette.py
```



the demo

a drawing of the app

switch to the other window now.

field notes, entry no. 19

What these notes *leave out.*

Contacts from Europe, 2005 to 2012. Partners from the US, 2017.

People who meet online. The diaries are too old to see them.

The guess for recognizing each other. Stated, never measured.

each one is a project for next year.

field notes, entry no. 20

Go talk *to people.*

About 14 of them could be the one.

your name / code: link to the repo



Data: POLYMOD, Mossong et al. 2008, Zenodo 3874557, CC BY-NC 4.0 / France 2012, Béraud et al. 2015, Zenodo 3886590, CC BY-NC-ND 4.0 / UN World Population Prospects 2024, CC BY 3.0 IGO / Rosenfeld, Michael J., Reuben J. Thomas, and Sonia Hausen. 2023. How Couples Meet and Stay Together 2017-2020-2022 combined dataset. Stanford University Libraries.